

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
sns.set(color_codes=True)
```

```
In [2]: ag = pd.read_csv("C:\\Users\\suryanm\\Downloads\\APY.csv\\APY.csv")
```

```
In [3]: ag
```

```
Out[3]:
```

	State	District	Crop	Crop_Year	Season	Area	Production	Yield
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	2439.6	3415.0	1.40
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	1626.4	2277.0	1.40
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	4147.0	3060.0	0.74
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	4147.0	2660.0	0.64
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	4153.0	3120.0	0.75
...	...	...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	855.0	1241.0	1.45
345332	West Bengal	PURULIA	Wheat	2016	Rabi	1366.0	2415.0	1.77
345333	West Bengal	PURULIA	Wheat	2017	Rabi	1052.0	2145.0	2.04
345334	West Bengal	PURULIA	Wheat	2018	Rabi	833.0	2114.0	2.54
345335	West Bengal	PURULIA	Wheat	2019	Rabi	516.0	931.0	1.80

345336 rows × 8 columns

```
In [4]: ag.shape
```

```
Out[4]: (345336, 8)
```

```
In [5]: ag.describe()
```

```
Out[5]:
```

	Crop_Year	Area	Production	Yield
count	345336.000000	3.453360e+05	3.403880e+05	345336.000000
mean	2008.887512	1.167147e+04	9.584726e+05	79.423135
std	6.564361	4.584079e+04	2.153068e+07	916.678396
min	1997.000000	4.000000e-03	0.000000e+00	0.000000
25%	2003.000000	7.400000e+01	8.700000e+01	0.550000
50%	2009.000000	5.320000e+02	7.170000e+02	1.000000
75%	2015.000000	4.112000e+03	7.182000e+03	2.470000
max	2020.000000	8.580100e+06	1.597800e+09	43958.330000

```
In [6]: ag.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 345336 entries, 0 to 345335
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   State           345336 non-null object
1   District        345336 non-null object
2   Crop            345327 non-null object
3   Crop_Year       345336 non-null int64
4   Season          345336 non-null object
5   Area            345336 non-null float64
6   Production      340388 non-null float64
7   Yield           345336 non-null float64
dtypes: float64(3), int64(1), object(4)
memory usage: 21.1+ MB
```

```
In [7]: ag.nunique()
```

```
Out[7]: State           37
District        707
Crop            55
Crop_Year       24
Season          6
Area            47712
Production      60410
Yield           10595
dtype: int64
```

```
In [8]: ag
```

```
Out[8]:
```

	State	District	Crop	Crop_Year	Season	Area	Production	Yield
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	2439.6	3415.0	1.40
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	1626.4	2277.0	1.40
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	4147.0	3060.0	0.74
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	4147.0	2660.0	0.64
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	4153.0	3120.0	0.75
...	...	...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	855.0	1241.0	1.45
345332	West Bengal	PURULIA	Wheat	2016	Rabi	1366.0	2415.0	1.77
345333	West Bengal	PURULIA	Wheat	2017	Rabi	1052.0	2145.0	2.04
345334	West Bengal	PURULIA	Wheat	2018	Rabi	833.0	2114.0	2.54
345335	West Bengal	PURULIA	Wheat	2019	Rabi	516.0	931.0	1.80

345336 rows × 8 columns

```
In [9]: ag['Season'].unique() ## the seasons of crop production in IND
```

```
Out[9]: array(['Kharif      ', 'Rabi      ', 'Autumn    ', 'Summer    ',  
              'Whole Year ', 'Winter    '], dtype=object)
```

```
In [10]: ag['Crop'].unique() ## the different type of crop in IND
```

```
Out[10]: array(['Arecanut', 'Arhar/Tur', 'Banana', 'Black pepper', 'Cashewnut',  
               'Coconut ', 'Cowpea(Lobia)', 'Dry chillies', 'Ginger', 'Groundnut',  
               'Maize', 'Moong(Green Gram)', 'Oilseeds total',  
               'Other Kharif pulses', 'other oilseeds', 'Rapeseed &Mustard',  
               'Rice', 'Sesamum', 'Sugarcane', 'Sunflower', 'Sweet potato',  
               'Tapioca', 'Turmeric', 'Urad', 'Bajra', 'Castor seed', 'Coriander',  
               'Cotton(lint)', 'Garlic', 'Gram', 'Guar seed', 'Horse-gram',  
               'Jowar', 'Linseed', 'Masoor', 'Mesta', 'Niger seed', 'Onion',  
               'Other Rabi pulses', 'Potato', 'Ragi', 'Safflower', 'Sannhamp',  
               'Small millets', 'Soyabean', 'Tobacco', 'Wheat',  
               'Peas & beans (Pulses)', 'Jute', 'Barley', 'Khesari', 'Moth',  
               'Other Cereals', 'Cardamom', 'Other Summer Pulses', nan],  
              dtype=object)
```

```
In [11]: ag.isnull().sum()
```

```
Out[11]: State          0
District        0
Crop            9
Crop_Year        0
Season           0
Area             0
Production      4948
Yield            0
dtype: int64
```

```
In [12]: ag.drop(['Production'],axis=1,inplace=True)
```

```
In [13]: ag
```

```
Out[13]:
```

	State	District	Crop	Crop_Year	Season	Area	Yield
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	2439.6	1.40
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	1626.4	1.40
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	4147.0	0.74
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	4147.0	0.64
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	4153.0	0.75
...	...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	855.0	1.45
345332	West Bengal	PURULIA	Wheat	2016	Rabi	1366.0	1.77
345333	West Bengal	PURULIA	Wheat	2017	Rabi	1052.0	2.04
345334	West Bengal	PURULIA	Wheat	2018	Rabi	833.0	2.54
345335	West Bengal	PURULIA	Wheat	2019	Rabi	516.0	1.80

345336 rows × 7 columns

```
In [14]: ag['Production'] = ag['Area ' ] * ag['Yield']
```

In [15]: ag

Out[15]:

	State	District	Crop	Crop_Year	Season	Area	Yield	Production
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	2439.6	1.40	3415.44
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	1626.4	1.40	2276.96
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	4147.0	0.74	3068.78
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	4147.0	0.64	2654.08
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	4153.0	0.75	3114.75
...	...	...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	855.0	1.45	1239.75
345332	West Bengal	PURULIA	Wheat	2016	Rabi	1366.0	1.77	2417.82
345333	West Bengal	PURULIA	Wheat	2017	Rabi	1052.0	2.04	2146.08
345334	West Bengal	PURULIA	Wheat	2018	Rabi	833.0	2.54	2115.82
345335	West Bengal	PURULIA	Wheat	2019	Rabi	516.0	1.80	928.80

345336 rows × 8 columns

In [16]: ag.isnull().sum()

Out[16]: State 0
District 0
Crop 9
Crop_Year 0
Season 0
Area 0
Yield 0
Production 0
dtype: int64

In [17]: gg = ag.drop(['Yield', 'Area'], axis=1)

```
In [18]: gg
```

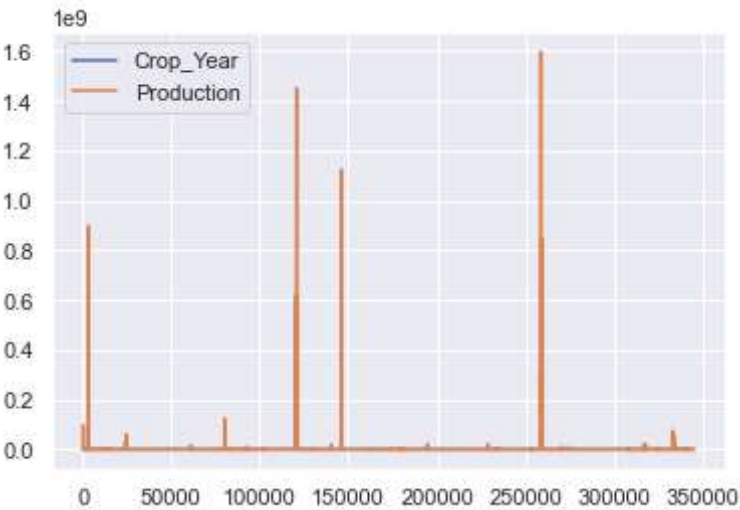
Out[18]:

	State	District	Crop	Crop_Year	Season	Production
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	3415.44
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	2276.96
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	3068.78
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	2654.08
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	3114.75
...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	1239.75
345332	West Bengal	PURULIA	Wheat	2016	Rabi	2417.82
345333	West Bengal	PURULIA	Wheat	2017	Rabi	2146.08
345334	West Bengal	PURULIA	Wheat	2018	Rabi	2115.82
345335	West Bengal	PURULIA	Wheat	2019	Rabi	928.80

345336 rows × 6 columns

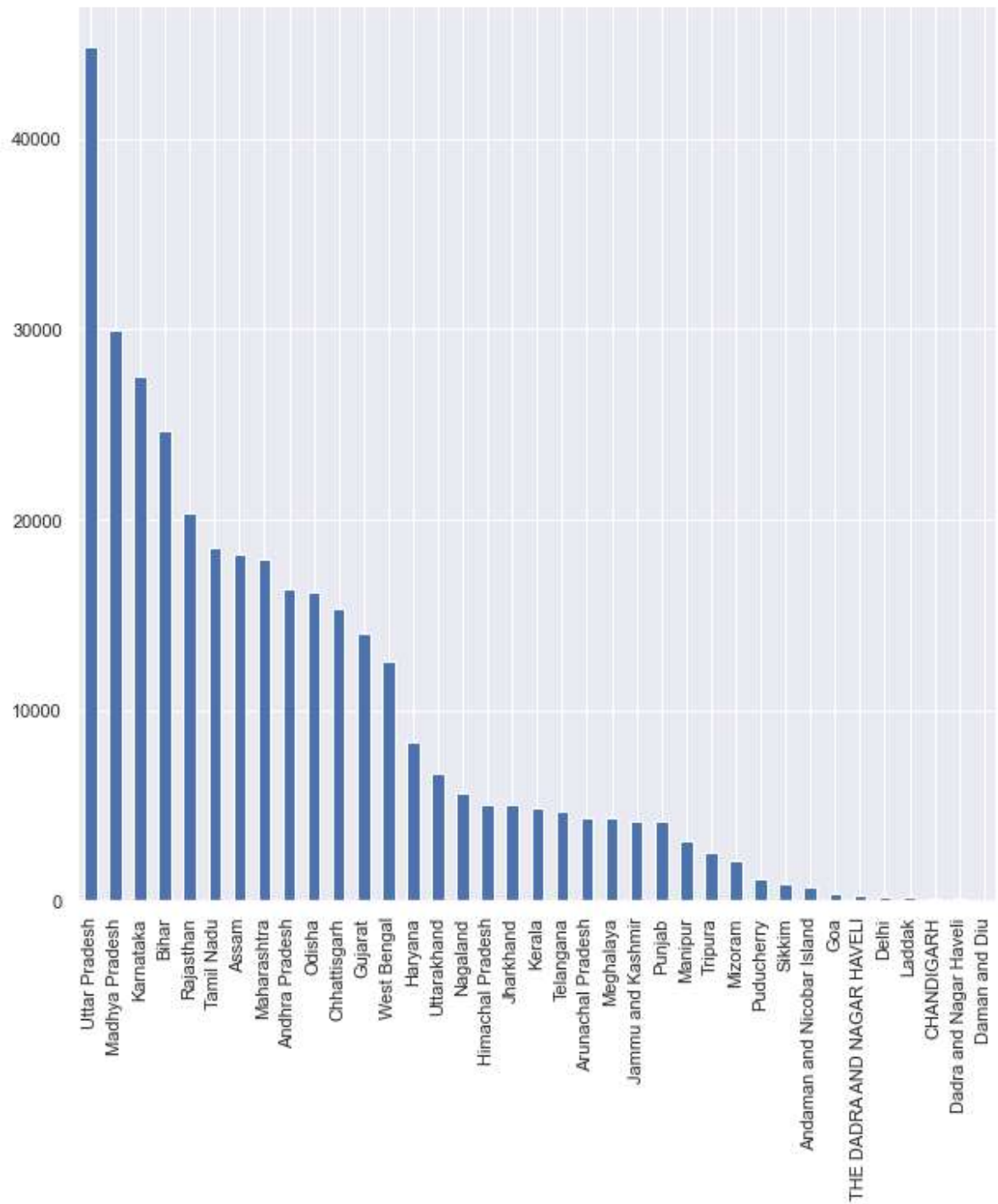
```
In [19]: gg.plot()
```

Out[19]: <AxesSubplot:>



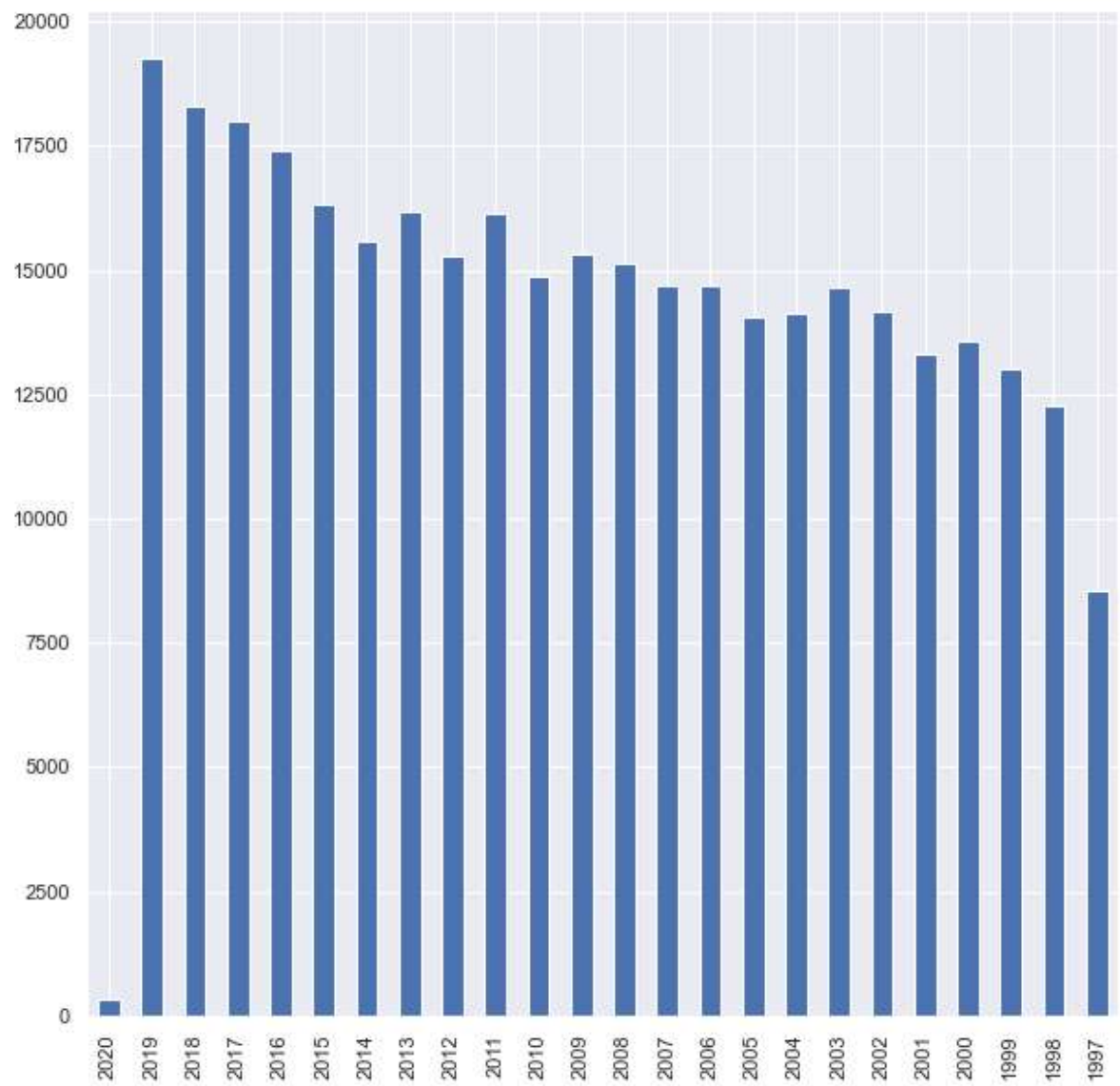
```
In [20]: gg.State.value_counts().plot(kind='bar',figsize=(10,10),)
```

```
Out[20]: <AxesSubplot:>
```



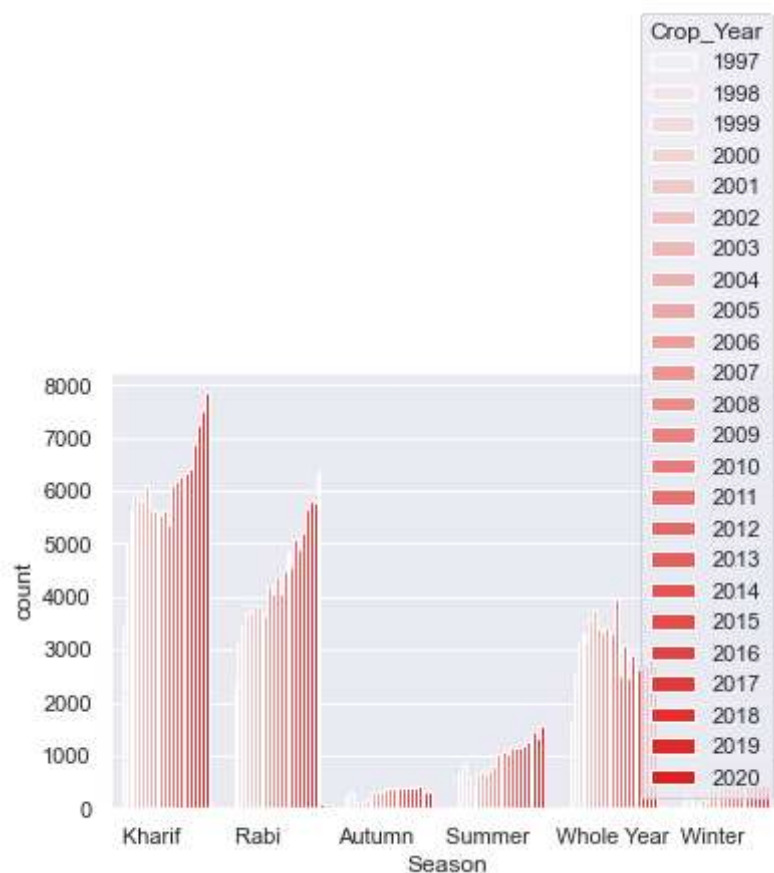
```
In [21]: gg.Crop_Year.value_counts().sort_index(ascending=False).plot(kind='bar',figsiz
```

```
Out[21]: <AxesSubplot:>
```



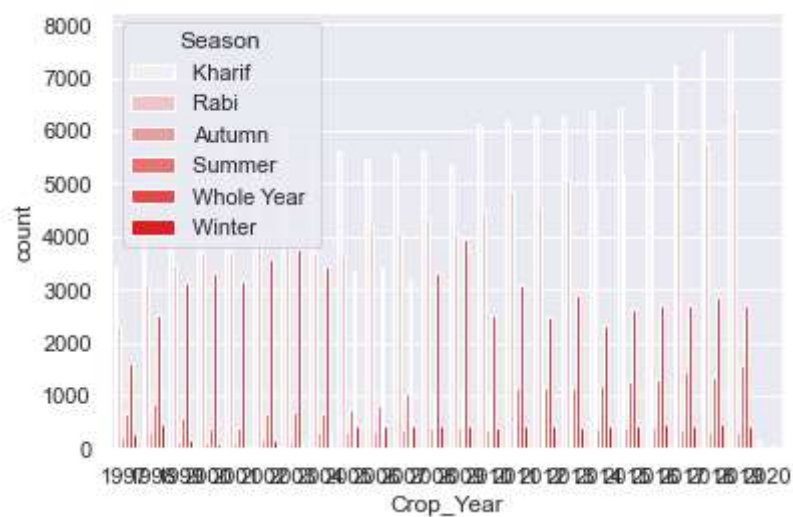

```
In [22]: sns.countplot(x='Season',data=gg,hue='Crop_Year',color='red')
```

```
Out[22]: <AxesSubplot:xlabel='Season', ylabel='count'>
```



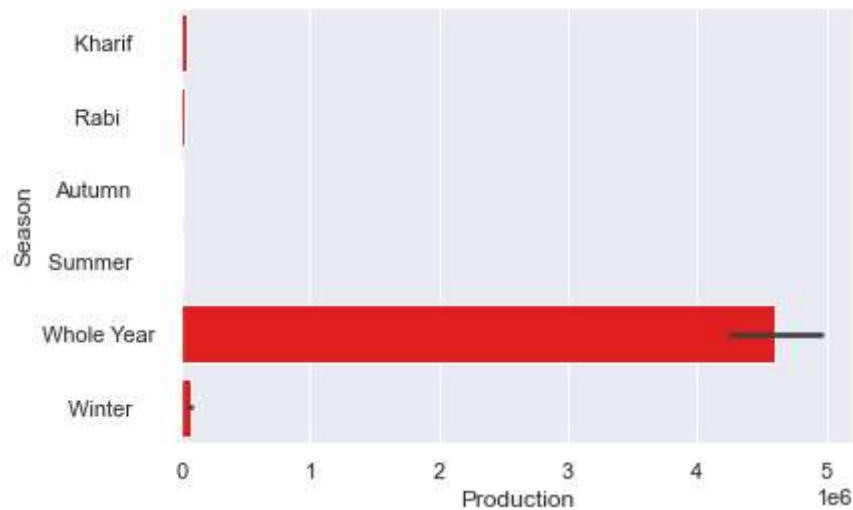
```
In [23]: sns.countplot(x='Crop_Year',data=gg,hue='Season',color='red')
```

```
Out[23]: <AxesSubplot:xlabel='Crop_Year', ylabel='count'>
```



```
In [24]: sns.barplot(x='Production',data=gg,y='Season',color='red')
```

```
Out[24]: <AxesSubplot:xlabel='Production', ylabel='Season'>
```



```
In [25]: sg=gg[(gg.Season.str.contains('Kharif'))|(gg.Season.str.contains('Winter'))|(g
```

```
In [26]: sg
```

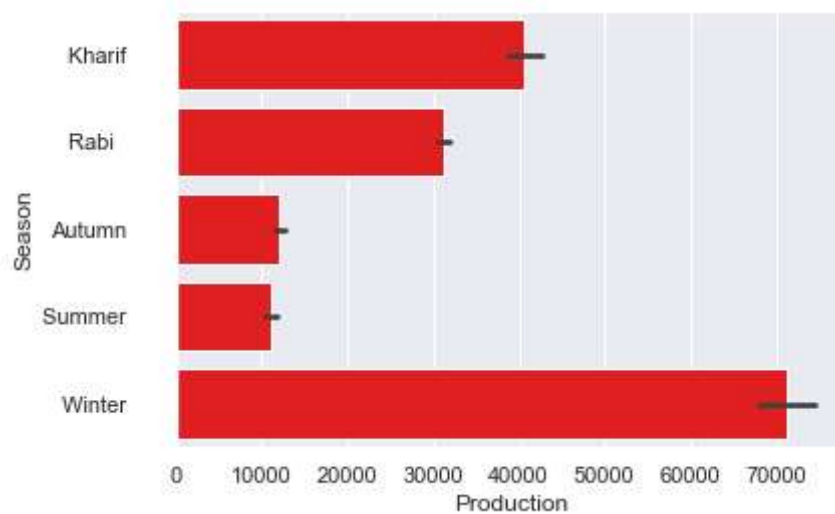
```
Out[26]:
```

	State	District	Crop	Crop_Year	Season	Production
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	3415.44
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	2276.96
2	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Autumn	3068.78
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	2654.08
4	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Autumn	3114.75
...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	1239.75
345332	West Bengal	PURULIA	Wheat	2016	Rabi	2417.82
345333	West Bengal	PURULIA	Wheat	2017	Rabi	2146.08
345334	West Bengal	PURULIA	Wheat	2018	Rabi	2115.82
345335	West Bengal	PURULIA	Wheat	2019	Rabi	928.80

276656 rows × 6 columns

```
In [27]: sns.barplot(x='Production',data=sg,y='Season',color='red')
```

```
Out[27]: <AxesSubplot:xlabel='Production', ylabel='Season'>
```



```
In [ ]:
```

```
In [28]: kh = gg[gg.Season.str.contains('Kharif')]
```

```
In [29]: kh
```

```
Out[29]:
```

	State	District	Crop	Crop_Year	Season	Production
0	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Kharif	3415.44
6	Andaman and Nicobar Island	NICOBARS	Arecanut	2000	Kharif	1993.86
7	Andaman and Nicobar Island	NICOBARS	Arecanut	2001	Kharif	2056.56
23	Andaman and Nicobar Island	NORTH AND MIDDLE ANDAMAN	Arecanut	2000	Kharif	5208.00
24	Andaman and Nicobar Island	NORTH AND MIDDLE ANDAMAN	Arecanut	2001	Kharif	5239.00
...	...	...	...	...	...	...
344903	West Bengal	PURULIA	Urad	2015	Kharif	2540.12
344905	West Bengal	PURULIA	Urad	2016	Kharif	5330.91
344907	West Bengal	PURULIA	Urad	2017	Kharif	5545.20
344909	West Bengal	PURULIA	Urad	2018	Kharif	3895.92
344910	West Bengal	PURULIA	Urad	2019	Kharif	5172.00

138369 rows × 6 columns

```
In [30]: kh.describe()
```

```
Out[30]:
```

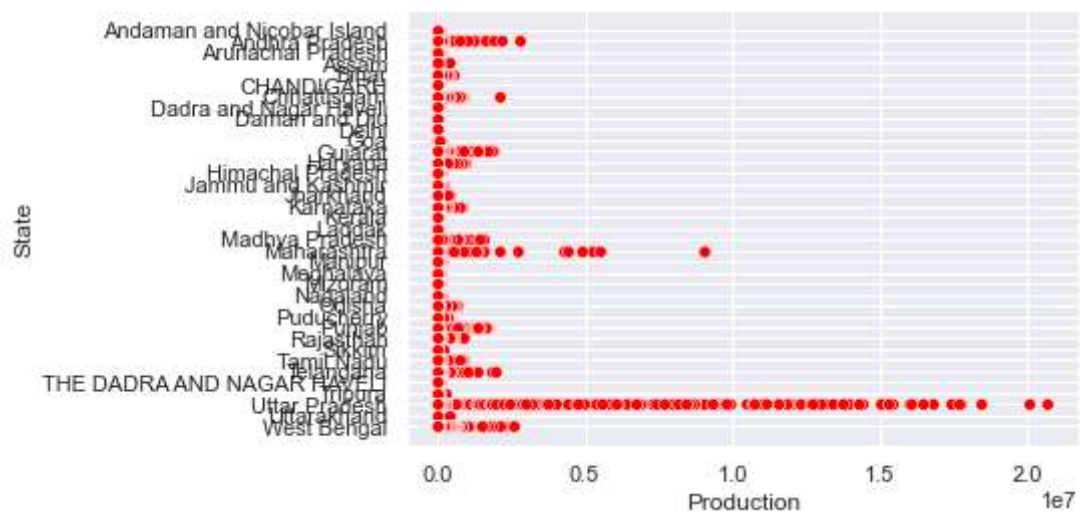
	Crop_Year	Production
count	138369.000000	1.383690e+05
mean	2008.800721	4.063356e+04
std	6.651707	3.761287e+05
min	1997.000000	0.000000e+00
25%	2003.000000	6.592000e+01
50%	2009.000000	5.978400e+02
75%	2015.000000	6.055200e+03
max	2020.000000	2.069354e+07

```
In [31]: kh.nunique()
```

```
Out[31]: State          37
District       698
Crop           53
Crop_Year      24
Season         1
Production     77702
dtype: int64
```

```
In [32]: sns.scatterplot(x='Production',data=kh,y='State',color='red')
```

```
Out[32]: <AxesSubplot:xlabel='Production', ylabel='State'>
```



```
In [33]: rh = gg[gg.Season.str.contains('Rabi')]
```

```
In [34]: rh
```

```
Out[34]:
```

	State	District	Crop	Crop_Year	Season	Production
1	Andaman and Nicobar Island	NICOBARS	Arecanut	2007	Rabi	2276.960
13	Andaman and Nicobar Island	NICOBARS	Arecanut	2010	Rabi	1614.240
14	Andaman and Nicobar Island	NICOBARS	Arecanut	2011	Rabi	1090.980
15	Andaman and Nicobar Island	NICOBARS	Arecanut	2012	Rabi	1361.780
16	Andaman and Nicobar Island	NICOBARS	Arecanut	2013	Rabi	845.975
...	...	...	...	...	...	...
345331	West Bengal	PURULIA	Wheat	2015	Rabi	1239.750
345332	West Bengal	PURULIA	Wheat	2016	Rabi	2417.820
345333	West Bengal	PURULIA	Wheat	2017	Rabi	2146.080
345334	West Bengal	PURULIA	Wheat	2018	Rabi	2115.820
345335	West Bengal	PURULIA	Wheat	2019	Rabi	928.800

100951 rows × 6 columns

```
In [35]: rh.describe()
```

```
Out[35]:
```

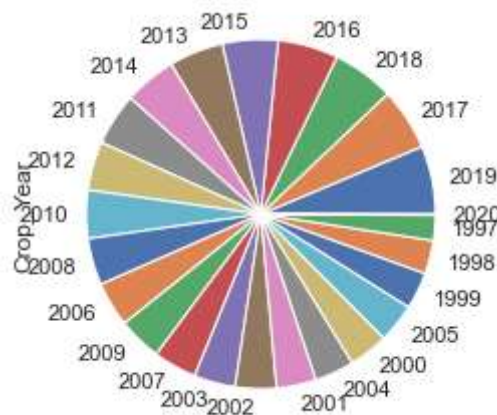
	Crop_Year	Production
count	100951.000000	1.009510e+05
mean	2009.385088	3.120816e+04
std	6.558282	1.287168e+05
min	1997.000000	0.000000e+00
25%	2004.000000	8.742000e+01
50%	2010.000000	7.027800e+02
75%	2015.000000	5.895300e+03
max	2020.000000	7.550488e+06

```
In [36]: rh.nunique()
```

```
Out[36]: State          36
District        698
Crop            52
Crop_Year       24
Season          1
Production      62689
dtype: int64
```

```
In [37]: rh.Crop_Year.value_counts().plot(kind='pie')
```

```
Out[37]: <AxesSubplot:ylabel='Crop_Year'>
```



```
In [38]: sm = gg[gg.Season.str.contains('Summer')]
```

```
In [39]: sm
```

```
Out[39]:
```

	State	District	Crop	Crop_Year	Season	Production
3	Andaman and Nicobar Island	NICOBARS	Arecanut	2008	Summer	2654.08
5	Andaman and Nicobar Island	NICOBARS	Arecanut	2009	Summer	2076.50
52	Andaman and Nicobar Island	NORTH AND MIDDLE ANDAMAN	Arhar/Tur	2008	Summer	91.70
53	Andaman and Nicobar Island	NORTH AND MIDDLE ANDAMAN	Arhar/Tur	2009	Summer	165.50
69	Andaman and Nicobar Island	NICOBARS	Banana	2008	Summer	7709.74
...	...	...	...	...	...	...
343018	West Bengal	PURULIA	Sesamum	2015	Summer	355.47
343021	West Bengal	PURULIA	Sesamum	2016	Summer	730.80
343024	West Bengal	PURULIA	Sesamum	2017	Summer	654.12
343027	West Bengal	PURULIA	Sesamum	2018	Summer	461.91
343029	West Bengal	PURULIA	Sesamum	2019	Summer	106.47

22098 rows × 6 columns

```
In [40]: sm.describe()
```

```
Out[40]:
```

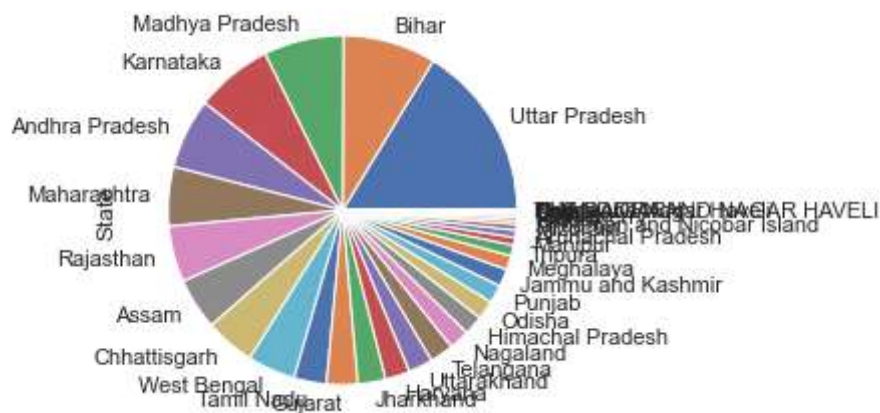
	Crop_Year	Production
count	22098.000000	22098.000000
mean	2010.062811	11031.103324
std	6.424024	48285.022312
min	1997.000000	0.000000
25%	2005.000000	67.867500
50%	2011.000000	487.275000
75%	2016.000000	3091.237500
max	2020.000000	883649.360000

```
In [41]: sm.nunique()
```

```
Out[41]: State          18
District       446
Crop           35
Crop_Year      24
Season          1
Production     15486
dtype: int64
```

```
In [42]: rh.State.value_counts().plot(kind='pie')
```

```
Out[42]: <AxesSubplot:ylabel='State'>
```



```
In [43]: win = gg[gg.Season.str.contains('Winter')]
```

```
In [44]: win
```

```
Out[44]:
```

	State	District	Crop	Crop_Year	Season	Production
32879	Assam	BAKSA	Rice	2005	Winter	93382.89
32882	Assam	BAKSA	Rice	2006	Winter	67853.50
32885	Assam	BAKSA	Rice	2007	Winter	73197.50
32888	Assam	BAKSA	Rice	2008	Winter	124396.05
32891	Assam	BAKSA	Rice	2009	Winter	110209.33
...	...	...	...	...	...	...
343013	West Bengal	PURULIA	Sesamum	2013	Winter	101.92
343016	West Bengal	PURULIA	Sesamum	2014	Winter	87.50
343019	West Bengal	PURULIA	Sesamum	2015	Winter	109.48
343022	West Bengal	PURULIA	Sesamum	2016	Winter	106.26
343025	West Bengal	PURULIA	Sesamum	2017	Winter	102.24

8249 rows × 6 columns

```
In [52]: win.describe()
```

```
Out[52]:
```

	Crop_Year	Production
count	8249.000000	8.249000e+03
mean	2009.585889	7.125448e+04
std	6.294439	1.593949e+05
min	1997.000000	0.000000e+00
25%	2005.000000	1.309500e+02
50%	2010.000000	3.425280e+03
75%	2015.000000	7.207326e+04
max	2019.000000	1.841754e+06

```
In [46]: win.nunique()
```

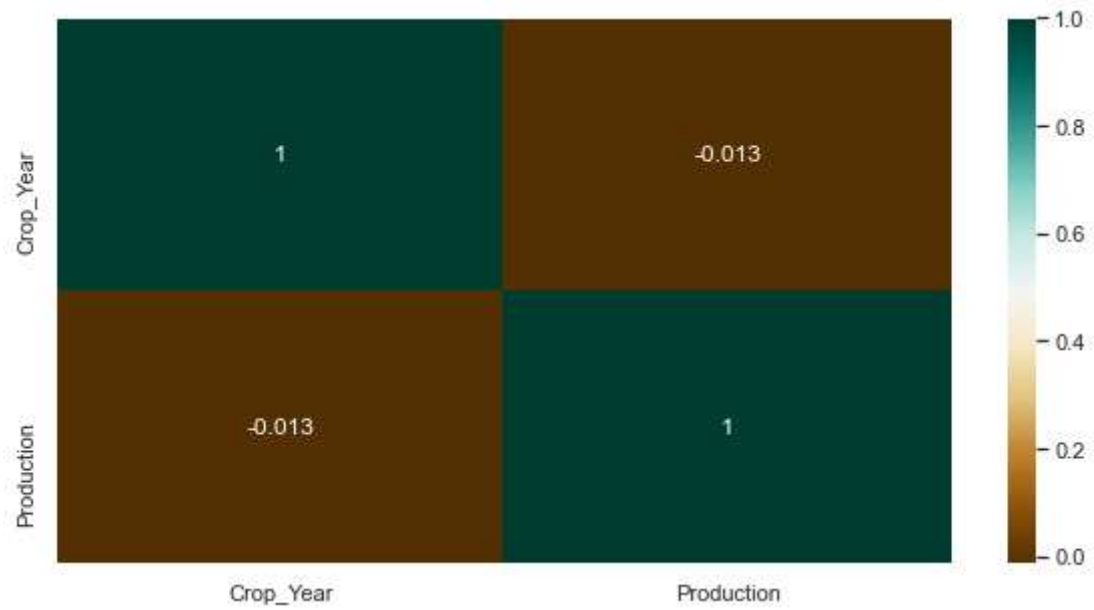
```
Out[46]: State      13
District    218
Crop        27
Crop_Year   23
Season      1
Production  7566
dtype: int64
```



```
In [51]: plt.figure(figsize=(10,5))
c=win.corr()
sns.heatmap(c,cmap="BrBG",annot=True)
c
```

Out[51]:

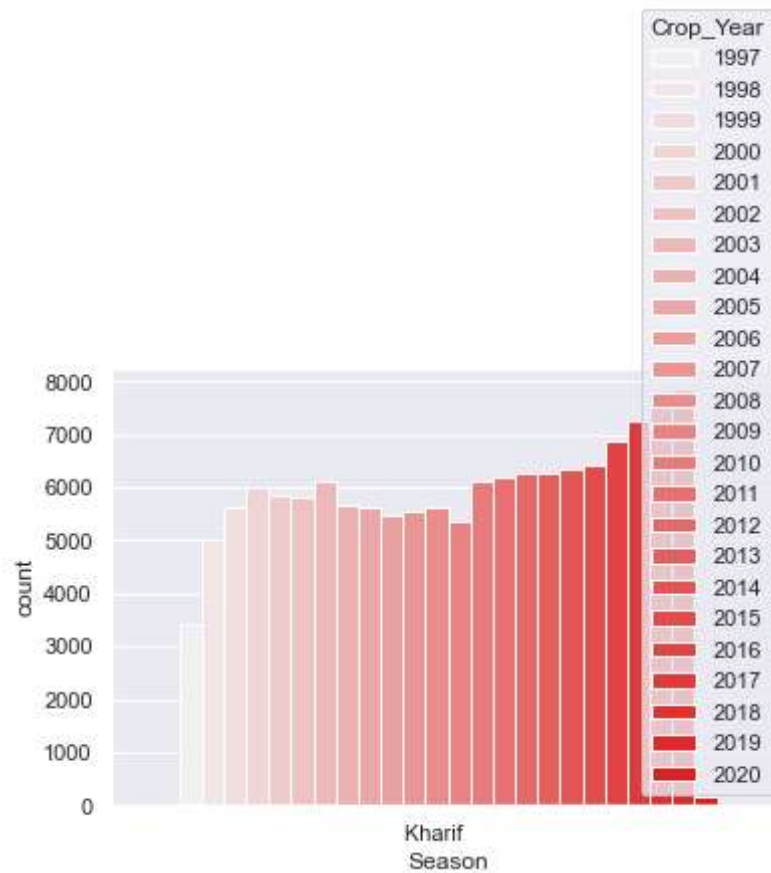
	Crop_Year	Production
Crop_Year	1.000000	-0.013332
Production	-0.013332	1.000000



In []:

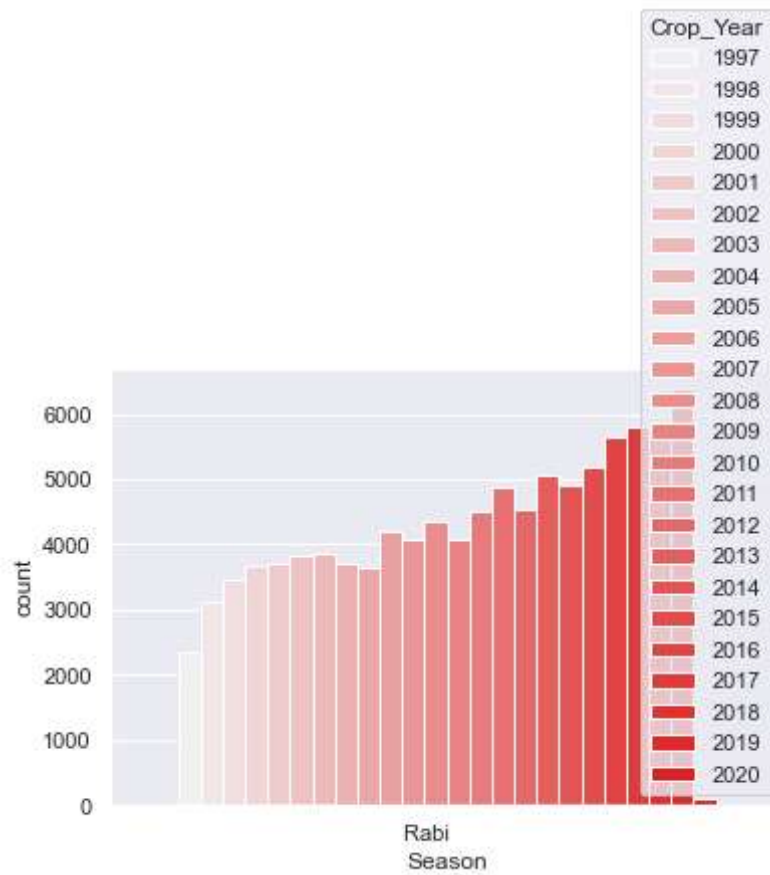
```
In [47]: sns.countplot(x='Season',data=kh,hue='Crop_Year',color='red')
```

```
Out[47]: <AxesSubplot:xlabel='Season', ylabel='count'>
```



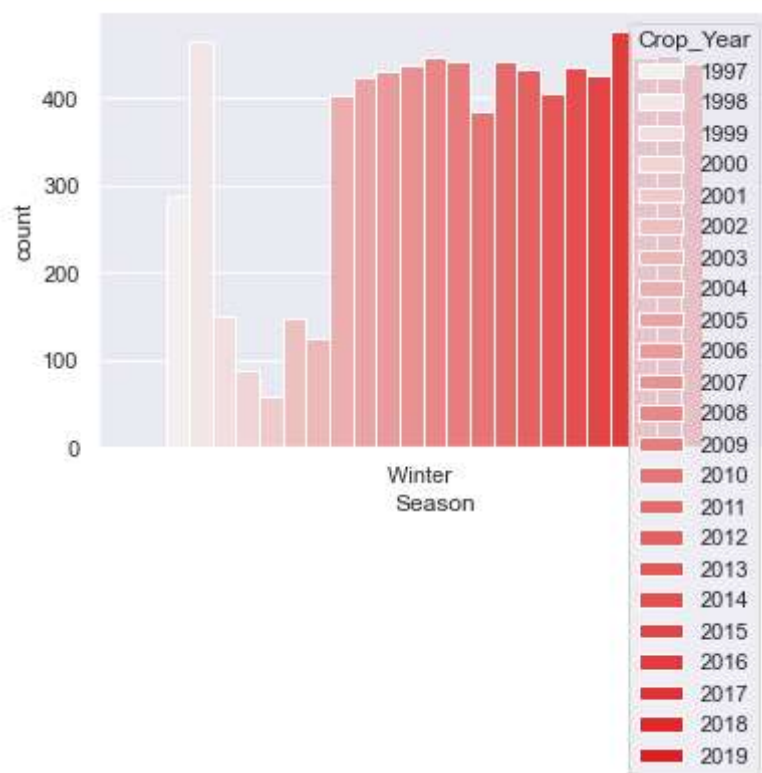
```
In [48]: sns.countplot(x='Season',data=rh,hue='Crop_Year',color='red')
```

```
Out[48]: <AxesSubplot:xlabel='Season', ylabel='count'>
```



```
In [49]: sns.countplot(x='Season',data=win,hue='Crop_Year',color='red')
```

```
Out[49]: <AxesSubplot:xlabel='Season', ylabel='count'>
```



```
Out[50]: <AxesSubplot:xlabel='Season', ylabel='count'>
```

